

Encapsulation in Java

Definition

Encapsulation is one of the fundamental principles of Object-Oriented Programming (OOP). It is the process of wrapping the data (variables) and methods (functions) that operate on the data into a single unit, typically a class.

Encapsulation helps to:

- **Protect the data** by making variables private.
- Provide **controlled access** through public methods (getters and setters).
- Achieve **modularity** and **code maintainability**.

Key Concepts

1. **Data Hiding:** The class variables are declared as private to prevent direct access from outside the class.
2. **Accessors (Getters):** Public methods to access private variables.
3. **Mutators (Setters):** Public methods to modify private variables.

Advantages

- **Improved Security:** Prevents unauthorized access to data.
- **Ease of Maintenance:** Changes in implementation do not affect external code.
- **Control Over Data:** You can validate data before setting it.

Code Example: Encapsulation with Student Class

```
// Student class demonstrating encapsulation
public class Student {
    // Private field
    private int rollNumber;

    // Getter method to access rollNumber
    public int getRollNumber() {
        return rollNumber;
    }

    // Setter method to set rollNumber
    public void setRollNumber(int rollNumber) {
        // Example of validation
        if (rollNumber > 0) {
            this.rollNumber = rollNumber;
        } else {
            System.out.println("Invalid roll number. Must be positive.")
        }
    }
}
```



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```
}  
}  
  
public static void main(String[] args) {  
    // Creating a Student object  
    Student student = new Student();  
  
    // Setting roll number using setter  
    student.setRollNumber(101);  
  
    // Getting roll number using getter  
    System.out.println("Roll Number: " + student.getRollNumber());  
}  
}
```

Output

Roll Number: 101

Interview Questions on Encapsulation

1. What is encapsulation in Java? Why is it important?
2. How do you achieve encapsulation in Java?
3. What is the difference between encapsulation and abstraction?
4. Why do we use private variables in encapsulation?
5. Can we achieve encapsulation without using getters and setters?
6. How does encapsulation improve security in Java?
7. Give a real-life analogy for encapsulation.
8. What happens if we don't use validation in setter methods?
9. Can encapsulation be achieved in languages other than Java?
10. How is encapsulation different from data hiding?